5 The .NET framework

Frequently, learning a language is less about syntax and more about getting a handle on the expansive libraries that the language provides. C# is no exception. Of course, having working knowledge of a library doesn't me know every method of every class. Rather, you should know the details of the components that you will use the most and know "the lay of the land" to find everything else quickly.

5.1 The layout of the .NET framework

First thing is where to look all this stuff up. For the MSDN documentation of the .NET framework, go to http://msdn.microsoft.com/en-us/library/ms229335.aspx. The .NET class libraries declare a hierarchy of namespaces that in turn contain the classes that we care about. At the top-level are two namespaces.

1. Microsoft which contains Microsoft-dependent technologies such as the support classes for the various Microsoft compilers, Windows-specific event handling, and tablet PC (ink) libraries.

2. System which contains the platform-agnostic technologies including the basic types, collections, IO, and networking. Our focus will be understanding the layout of System and some of the important classes in there.

5.2 The system namespace

At the root of the System namespace exists many important, general-purpose classes that you should know. These include

- Classes and structs corresponding to fundamental types (e.g., Int32, Char). These classes frequently contain static helper methods that help your process and convert data of those types.
- The Console class contains methods to manipulate the console. This goes beyond simple reading and writing. You can also manipulate the location of the cursor and other things.
- The Random class for random number generation.
- The Exception class and the basic exceptions that inherit from it. Note that unlike Java there is no notion of checked and unchecked exceptions. Checked exceptions must be caught by a callee or the enclosing method must declare that it throws a checked exception by a throws clause. All exceptions are unchecked (i.e., may not be caught) in C#.

```
public static class NumberConversionExample
```